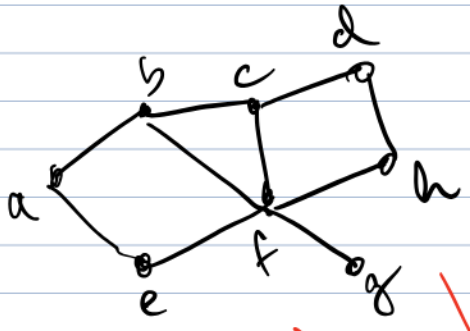
Graph Theory

SUBRAHMANYAM KALYANA-SUNDARAM.

Already defined : Graphs, degree, min & max degree, neighbor, subgraph, induced subgraph. paths, cycles.



$d_G(a,h) = 3$. → $\text{dia}(G) = 3$.

Already seen : Every graph contains a path of length $\delta(G)$ and cycle of length $\geq \delta(G)+1$. (look at the longest path).

Distance: The distance $d_G(x,y)$ of two vertices the length of a shortest x-y path in G. If no such path exists $\} \; d_G(x,y) = \infty$

Diameter of G: $\max_{x,y \in V(G)} d(x,y)$

$\text{diam}(G)$.

Girth: The minimum length of a cycle in a graph is the girth $g(G)$ of G.
The maximum length of a cycle is called circumference of G. In simple graphs G, $g(G) \geq 3$.

If G is acyclic, then $g(G) = \infty$
circumference $= 0$

A vertex is central in G if its greatest distance from any other vertex is as small as possible. The distance is called the radius.

Radius $\operatorname{rad}(G) = \min_{x \in V(G)} \max_{y \in V(G)} d_G(x,y)$

$$\operatorname{Rad}(G) \leq \operatorname{diam}(G) \leq 2 \operatorname{Rad}(G).$$

Why? Fix $u \in V(G)$.

$$\operatorname{Rad}(G) \leq \max_{y \in V(G)} d_G(u,y)$$

$$\max_{y \in V(G)} d_G(u,y) \leq \max_{x,y \in V(G)} d_G(x,y)$$

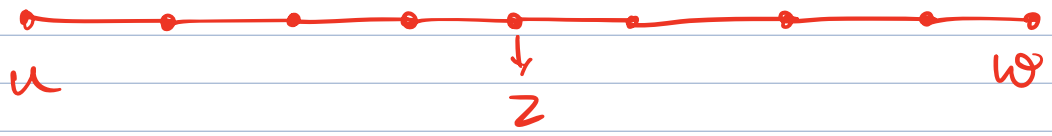
$$\| $$

$$\text{diam}(G).$$

Why is $\text{diam}(G) \leq 2\,\text{Rad}(G)$?

Let $u, w \in V(G)$ be the vertices that are the farthest apart.

That is $\text{diam}(G) = \max_{x,y \in V(G)} d(x,y) = d(u,w)$



Let z be a mid point on this shortest u-w path.

For z, $\max_{y \in V(G)} d_G(z,y) \geq \frac{d_G(u,w)}{2}$

$$= \text{diam}(G)/2.$$

Rad (G) cannot be smaller than diam $(G)/2$

Suppose it is smaller. Rad $(G) <$ diam $(G)/2$

There exists a point t such that

$d_G(u,t)$, $d_G(w,t) <$ diam $(G)/2$.

$\Rightarrow d_G(u,w) <$ diam (G).

A contradiction.

Prop 1.3.3: A graph G of radius $\leq k$ and max degree $\leq d$ (where $d \geq 3$) has $\leq \frac{d}{d-2}(d-1)^k$ vertices.

Proof: Let z be a central vertex of G. Let D_i be the set of vertices at a distance i from z.

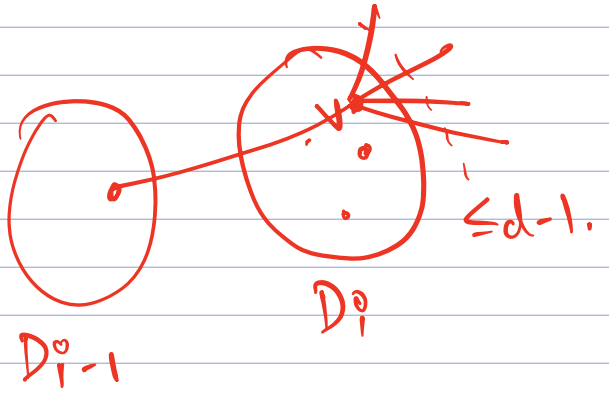
Clearly $V(G) = \bigcup_{i=0}^{k} D_i$.

$D_0 = \{z\}$.

D_1 = set of neighbors of z.

$|D_1| = \deg(z) \leq d$.

For $i \geq 1$, $|D_{i+1}| \leq (d-1)\,|D_i|$.



Every vertex in D_{i+1} is a neighbor of a vertex in D_i.

Every vertex in D_i has $\leq (d-1)$ neighbors in D_{i+1}

So $|D_{i+1}| \leq (d-1)^i d$. for all $i < k$.

$$|V(G)| = 1 + \sum_{i=1}^{k} |D_i|$$

$$\leq 1 + \sum_{i=1}^{k} d\,(d-1)^{i-1}$$

$$= 1 + d \sum_{i=0}^{k-1} (d-1)^i$$

$$= 1 + d \cdot \frac{(d-1)^k - 1}{(d-1) - 1}$$

$$< 1 + \frac{d}{d-2} (d-1)^k .$$

Section 1.4 : Connectivity

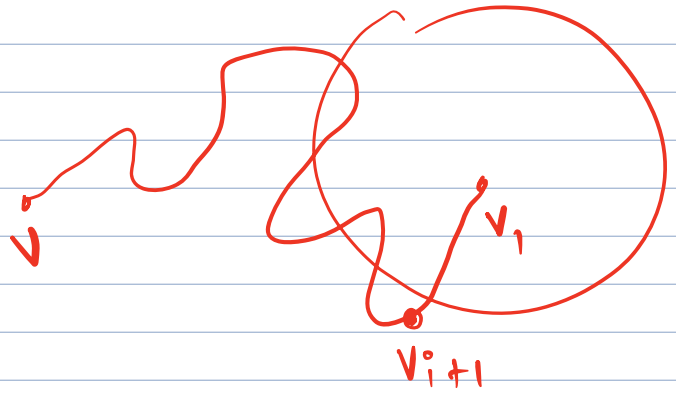
A graph G is connected if for any two vertices $u, v \in V(G)$, there is a u-v path in G. If $U \subseteq V(G)$, and $G[U]$ is connected, we also say U is connected.

Prop 1.4.1 : The vertices of a connected graph G can be ordered $v_1, v_2, \dots v_n$ so that $G[v_1, v_2 \dots v_i]$ is connected for any i.

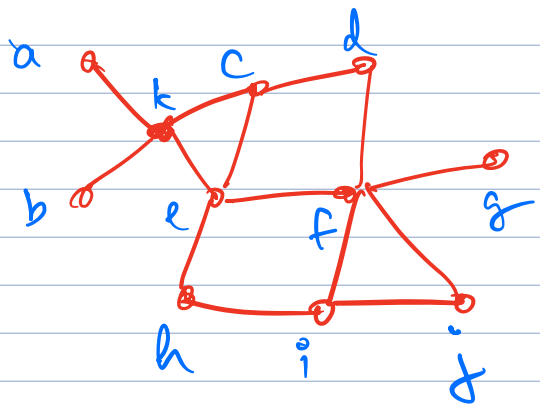
Proof : Base case : Set any vertex as v_1.

Let $v_1, v_2 \dots v_i$ be such that $G[v_1, \dots v_i]$ is connected. Let $G_i = G[v_1, v_2 \dots v_i]$.

Let $v \in G - G_i$. As G is connected, there is a path from v to v_1.



Pick v_{i+1} as the last vertex in this path that is not in G_i. v_{i+1} has a neighbor in G_i as it enters G_i in the next step. So $G_i \cup \{v_{i+1}\} = G[v_1, v_2 \cdots v_i, v_{i+1}]$ is also connected.



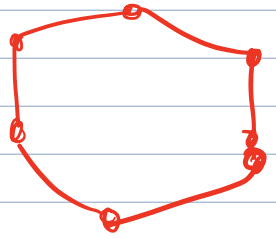
girth = 3

circumference = 8

diameter = 4

Central vertex = e is the only central vertex

Radius = 2.



→ Every vertex is central.

Radius = 3.

$$\text{Radius}(G) \leq \text{Diam}(G) \leq 2\,\text{Radius}(G).$$

Cycle above has diam = radius = 3

$\Leftarrow$ ~~Radius = $\lceil \frac{\text{Diam}}{2} \rceil$~~

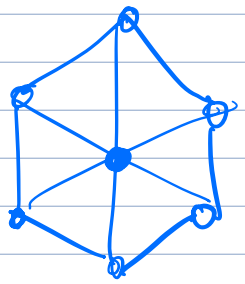


Path on 6 vertices

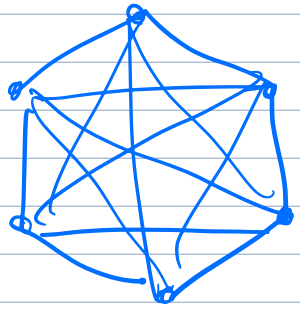
Diam = 5

Radius = 3

There are 2 central vertices



Diameter = 2
Radius = 1
There is only one central vertex

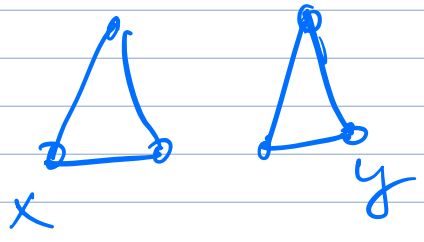


→ K_6 : 6 vertices and every pair of vertices are adjacent.

True for any complete graph.

{ Diameter = Radius = 1
Every vertex is central

- Single vertex graph } I think diam = rad = 0



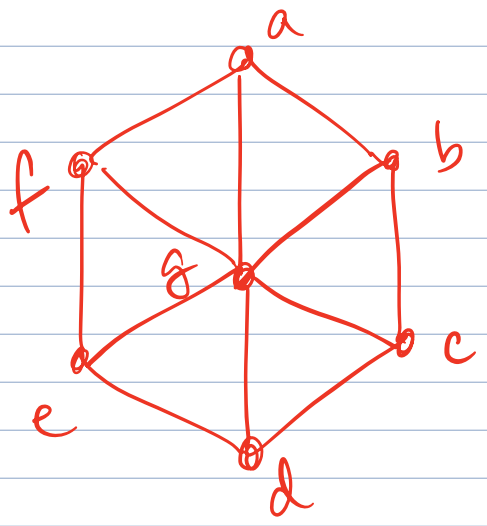
central

Q. Given G, are all the vertices of G part of x-y path in G, where x-y determine the diameter?

Q: w is central vertex $\iff$ w lies in the middle of some x-y path with $d(x,y)$ = diam

In fact both directions.

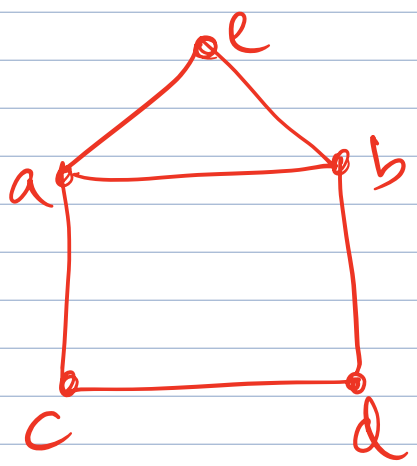
This statement is wrong. Consider below graph G.



Diameter = 2

Consider the path a-b-c

The middle vertex here is b which is not a central vertex.



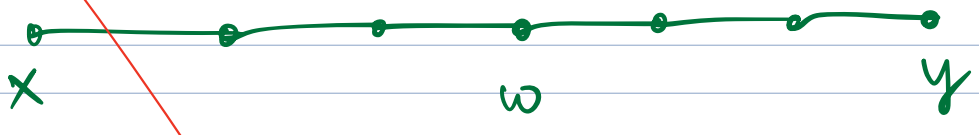
Vertex e is central
In fact every vertex is central, and radius = 2.

But e is not a middle vertex of any shortest path. In fact, diam = 2 as well.

Exercise 1: The statements (both directions are true for trees.

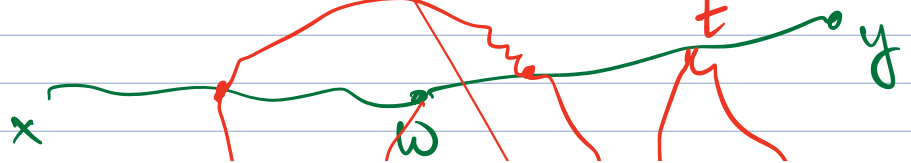
Exercise 2: Find the wrong step in our proof attempt.

($\Leftarrow$) Suppose w lies in the middle of a shortest x-y path. $d(x,w), d(y,w) \leq \left\lceil \frac{d(x,y)}{2} \right\rceil$



Claim: For all $z \in V(G)$, $d(w,z) \leq$ radius

Proof: Suppose $\exists$ z such that $d(w,z) >$ radius





length of red w-t path ≥ length of green w-t path

Otherwise it contradicts the assumption that x-y is a shortest path ⇒ any subpath in the x-y path is also a shortest path.

Combining the green w-t path + the t-z path we get a path that is longer than w-y path. Then x-w-t-z is a longer shortest path than x-y, contradicting the choice of x-y.

So we have $\max_{z \in V(G)} d(w,z) = \text{radius}$.

So w is central.

(⇒) Suppose w is a central vertex.

Consider any $x, y \in V(G)$ such that

$$d(x,y) = \text{diam}(G).$$

Claim: w lies in the middle of a shortest x–y path.

Proof: Since w is a central vertex

$$d(x,w),\ d(y,w) \leq \text{radius}$$

Consider the $x-w-y$ path obtained by merging a shortest x-w path with a shortest w-y path.

This path has length $\leq 2 \cdot \text{radius}$.